Testing SWINGERsim

Part IV: Special cases – the future of CzeV343, the hierarchy of CzeV4315 and the convergence of the posteriors

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1 Aim

Parts I–III of this test series asked whether SWINGERsim reproduces what is already known: published benchmarks of the physics (Part I), three well-observed multiple stars (Part II) and three eclipsing binaries whose inclination changes because of a third body (Part III). Part IV asks questions to which no published answer exists. These are the questions for which the software was built.

1. **How and when does CzeV343 fall apart?** CzeV343 is a doubly eclipsing 2+2 quadruple. Pejcha et al. (2022) found that the quadruple becomes dynamically unstable if one of its binaries widens beyond about 50 $R_\odot$ by mass transfer. They did not say when that happens or what follows (“dynamical instability caused by orbital expansion … has not been explored so far in 2+2 quadruples”). We follow the stellar evolution

and mass transfer of both binaries to the onset of the instability. We then integrate the four-body motion from that moment on (Section 2).

2. **Is CzeV4315 a 2+2 quadruple or a 2+(2+1) quintuple?** In CzeV4315 the inclination of binary B changes by 1.4° per year (Zasche et al. 2025). B also shows a 119-day cycle in its eclipse times. We compare two hierarchies with the same data, weigh them by the Bayesian evidence, and ask what the preferred one predicts (Section 3).
3. **Are the posteriors of the fits converged?** The Markov-chain Monte Carlo (MCMC) fits of Parts II and III were run for 1000 steps. We check how far they are from convergence and what this means for the quoted uncertainties (Section 4).

A word on the two kinds of result in this part. A **forward model** starts from the present system and integrates it into the future; its answer is a distribution over the unknown parameters, not a single date. An **ensemble** is a set of such models whose unknown parameters are drawn from stated prior ranges. Each member gets an **observation weight**: the likelihood of the observed data given that member. The mean weight over the ensemble estimates the **Bayesian evidence** Z of the hypothesis behind the ensemble; the ratio of two evidences, the **Bayes factor**, says how much better one hypothesis explains the data than the other. The **effective sample size** $n_{\mathrm{eff}} = (\sum w)^2 / \sum w^2$ counts how many members carry the result. A result resting on a handful of members is a hint, not a measurement. The methods are described in *Physics and Methods*, chapters Ensembles and inference and Fitting observed data.

All numbers in this part come from the output files listed in Data and reproducibility. The figures are made by `tools/part4_figures.py`.

2 CzeV343: when and how does the quadruple fall apart?

2.1 The system and the question

CzeV343 (V849 Aur) is a doubly eclipsing quadruple. Two eclipsing binaries orbit each other (Pejcha et al. 2022), so the hierarchy is 2+2:

- **Binary A**: period 1.2093 d, eccentricity 0.114, apsidal period 33 yr.
- **Binary B**: period 0.8069 d, circular. The ratio of the two periods is close to 3:2.
- **Mutual orbit of A and B**: period 1454 d (4.0 yr), eccentricity 0.70, inclined only about 12° to the plane of the sky. Both binaries are seen nearly edge-on, so their orbital planes are strongly misaligned with the mutual orbit: 76–104° for A, and 63–86° or 94–117° for B.

Pejcha et al. fitted isochrones to the light of the four stars and gave two solutions (their Table 2):

- **Fiducial**: age about 630 Myr; A: 1.76 + 1.27 $M_\odot$; B: 1.36 + 1.20 $M_\odot$.
- **Alternative**: age about 140 Myr; A: 2.48 + 1.42 $M_\odot$; B: 1.75 + 1.38 $M_\odot$.

They followed both binaries with the binary population synthesis code COMPAS (Riley et al. 2022), whose stellar and binary prescriptions descend from SSE/BSE (Hurley et al. 2000, 2002), on a grid around the measured parameters. In $\gtrsim 85$ % of their realisations binary A widens significantly by mass transfer, to a largest separation of 50–220 $R_\odot$; about 60 % of the realisations of binary B widen to 80–150 $R_\odot$. Most of these binaries later merge; about 20 % of A survive as double white dwarfs. Pejcha et al. applied the triple stability criterion of Mardling & Aarseth (2001) to each binary, with the other binary as a point mass. They found that the quadruple stays dynamically stable only while $a_A, a_B \lesssim 50$ $R_\odot$, and concluded that "essentially all realizations of binary A that survive main sequence as a binary will experience this dynamical instability". They did not follow the system further: "This instability has not been explored so far in 2+2 quadruples".

Our question has two parts: **when** does the instability start, and **what happens then** – ejection of a star, a collision, an exchange of partners, or a new stable configuration?

2.2 Method: two stages

Integrating four stars directly over 2 Gyr with orbital periods of one day is not feasible. The problem separates into two timescales, so we split it into two stages.

Stage 1 – stellar evolution and mass transfer (millions of years). We take 300 draws for each of the two isochrone solutions. In each draw:

- the mass of each binary varies by 10 % (Gaussian σ; Pejcha et al. estimate an uncertainty of about 10 % for the stellar masses from the choice of metallicity);
- the mass ratio within each binary varies by ±5 %;
- the age is multiplied by a log-normal factor (σ = 0.3 in ln age), because isochrone ages are uncertain;
- the orientations (nodes) of the two binaries are drawn at random. The nodes fix the mutual inclinations, which the stability criteria depend on.

For each draw SWINGERsim computes the mass-transfer history of both binaries up to 4 Gyr from now: stellar tracks (MIST), Roche-lobe overflow, the stability and rate of the transfer, contact and common envelope (*Physics and Methods*, chapter Mass transfer and evolution). Along each history it evaluates the two stability criteria of both nested triples – each binary with the other binary as a point mass:

- **Mardling & Aarseth (2001) criterion (MA01)**: a ratio of the outer periastron to the critical value. Below 1 the triple is unstable.
- **Vynatheya et al. (2022) criterion (V22)**: a fit to many N-body integrations, with the same meaning of "below 1".

The outer orbit follows the masses adiabatically ($a_{\rm out}M_{\rm tot}$ = const for isotropic mass loss). The first time either ratio drops below 1 is the **onset of instability**.

A draw that does not finish within 300 s of computing time, usually because the stellar track leaves the table, is recorded as unfinished. This affected 12 fiducial and 14 alternative draws (4–5 %). We see no reason why these draws should differ systematically from the others.

Stage 2 – dynamics after the onset (thousands of years). For six onsets we set up a direct N-body integration at that moment. These are five fiducial onsets spread over the onset times and the one alternative onset that reached N-body. The binary that widened (always A) stays a binary, with the masses, separation, radii and temperatures of its history. The other binary, B, is 16–20 times tighter than the outer periastron ($a_B/a_{\rm out} \approx 0.005$), so we treat it as a single body. Its "radius" is its separation, so that a close approach to B counts as a collision. General relativity, tides and stellar rotation are switched on. Each onset is integrated for 10^4 yr (about 2500 outer orbits) as an ensemble of 100 members. The members differ in the orbital phases of both orbits and in the argument of periastron of A, which are not determined by the evolution.

One quantity in stage 2 needs separate treatment. The donor of binary A is, at the onset, a **stripped giant**: 0.26–0.30 $M_\odot$, 13.4–14.5 $R_\odot$, 3850–3980 K. It has lost almost all of its envelope to its companion, which now has 2.6–3.0 $M_\odot$. How strongly such a star is distorted by its companion depends on its internal mass concentration, measured by the **apsidal-motion constant** k_2. A star with all its mass at the centre has $k_2 = 0$, a uniform sphere has 0.75, and main-sequence stars have about 0.005–0.02. The tidal and rotational bulges make the orbit of A precess (apsidal motion), and this precession competes with the gravitational pull of B (see below). The value of k_2 for this star is not in any table, so we ran every onset with three values: 0.001, 0.01 and 0.05. That makes 18 ensembles with 1800 members in total.

What is SWINGERsim's own work here. These parts were programmed for SWINGERsim:

- the mass-transfer histories: stability of the transfer, rates, contact, common envelope, and the coupling to the stellar tracks;
- the evaluation of both stability criteria along the histories;
- the construction of the N-body state at the onset;
- the ensemble machinery and its statistics;
- the apsidal-constant model of the next section.

The stellar tracks come from MIST (Choi et al. 2016). The N-body integration uses the IAS15 integrator of REBOUND (Rein & Spiegel 2015), with relativity (`gr_full`) and the self-consistent tides and spins (`tides_spin`; Lu et al. 2023) of REBOUNDx (Tamayo et al. 2020).

2.3 Stage 1: what happens to the two binaries

Table 1. Outcome of stage 1 for binary A (the heavier binary, which interacts first). Each row is one isochrone solution. The columns give the share of finished draws whose binary A ends in each state:

- *merger*: contact, then a common envelope, then the two stars merge;
- *contact*: a contact binary at the end of the computation;
- *stripped*: the donor loses its envelope and the orbit widens;
- *none*: no interaction within 4 Gyr.

The next two columns give the share of draws in which A widens beyond 50 $R_\odot$, and the share that reaches the dynamical instability. The last column gives the onset times (16th, 50th and 84th percentiles).

solution	draws	merger	contact	stripped	none	$a_A >$ 50 $R_\odot$	unstable	onset of instability [Gyr from now]
fiducial	288	59 %	30 %	10 %	1 %	29 %	14 %	1.30 – 1.49 – 1.63
alternative	286	0 %	99 %	1 %	0 %	3 %	1 %	1.85 – 1.95 – 2.04

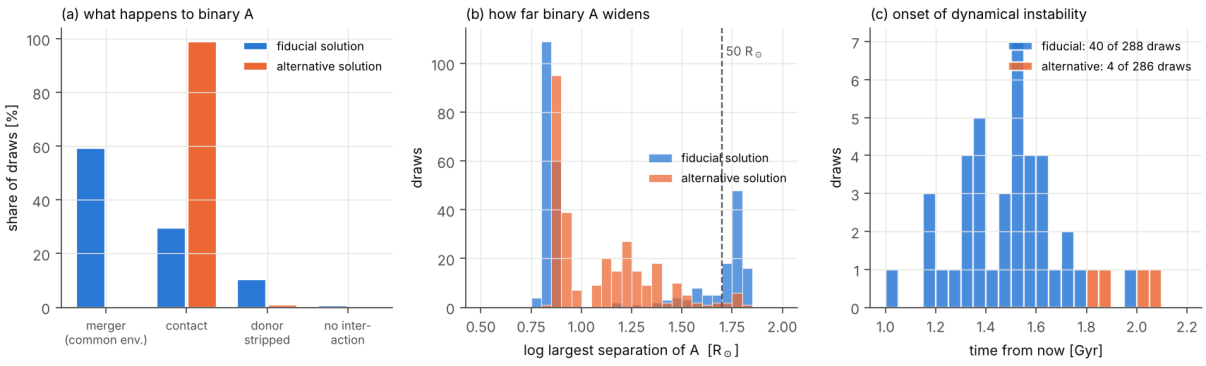


Figure 1. Stage 1 of CzeV343 for the fiducial (blue) and alternative (orange) isochrone solution. **(a)** What happens to binary A, as a percentage of the draws: merger after a common envelope, contact, stripped donor with a widened orbit, or no interaction. **(b)** Histogram of the largest separation that binary A reaches in each draw (logarithmic axis, $R_\odot$). The dashed line marks 50 $R_\odot$, above which Pejcha et al. (2022) expect the quadruple to become unstable. **(c)** When the quadruple becomes dynamically unstable (Gyr from now), for the draws in which it does; the legend gives how many draws these are.

Figure 1 and Table 1 show the outcome of stage 1.

Binary A almost always interacts. Its heavier star (1.76 $M_\odot$ in the fiducial solution) fills its Roche lobe while it still burns hydrogen in its core. This is **case A** mass transfer, typical of periods of about a day.

- In the **fiducial** solution the transfer most often brings the two stars into contact. The binary then merges after a common envelope (59 %, median 1.4 Gyr from now, 16–84 %: 1.0–2.0 Gyr) or remains a contact binary to the end of the computation (30 %).
- In the **alternative** solution the stars are more massive and younger, and binary A becomes a contact binary in 99 % of draws.

In both cases the separation stays small: in panel (b) the tall peaks near $\log a \approx 0.85$ (7 $R_\odot$) are these draws.

Only a stripped donor widens the orbit. In 10 % of fiducial draws the donor loses its whole envelope before contact; these draws have the heaviest donors ($\gtrsim 1.8$ $M_\odot$). The transfer then reverses the mass ratio and the orbit widens to 53–66 $R_\odot$: the second peak in panel (b), right of the dashed line. These draws, and a few that reach contact only after widening to above 50 $R_\odot$, are the 29 % beyond 50 $R_\odot$.

Half of the draws beyond 50 $R_\odot$ (14 % of all) cross the stability limit. The other half do not, because the critical separation depends on the mutual inclination and on the outer eccentricity at that moment. The onsets fall between 1.0 and 2.1 Gyr from now (panel c); the fiducial median is 1.49 Gyr. At the onset binary A has a separation of 57–62 $R_\odot$. It is always binary A that triggers the instability: binary B becomes a contact binary (fiducial 70 %, alternative 95 %) without ever widening.

The two criteria disagree at the onset. By construction MA01 equals 1.00 at the onset, but V22 is still 1.19–1.24, so by Vynatheya et al. the system is still stable. The criteria were calibrated on different sets of integrations and differ by about 20 % near the boundary. This is why stage 2 is needed: the N-body integration decides, not the criterion.

Which criterion to trust here. A systematic test answers this for CzeV343 after the mass transfer: binary A of 2.77 + 0.26 $M_\odot$ with periods of 12–80 d, binary B of 1.36 + 1.20 $M_\odot$ with 4 d, outer eccentricity 0.5–0.8, isotropic orientations, chosen so that the MA01 ratio lies between 0.6 and 1.6. All 1200 members were integrated for 200 outer orbits (800 yr) with MEGNO. The 243 members that stayed intact but were chaotic or below the MA01 limit were integrated for 10^4 outer orbits (4×10^4 yr). In addition, the ghost-orbit test of Vynatheya et al. (2023), on which their neural network was trained, was run for every member. Stars are point masses with small radii (0.05 $R_\odot$), so ZLK collisions are recorded but are not counted as dynamical break-ups.

Table 2b. CzeV343 after the mass transfer: share of the members that break up (ejection, exchange or loss of the hierarchy) by the MA01 ratio at the start.

MA01 ratio	0.6–0.7	0.7–0.8	0.8–0.9	0.9–1.0	1.0–1.1	1.1–1.6
members	121	142	126	157	171	483
break-up within 800 yr	69 %	39 %	18 %	5.7 %	1.2 %	0 %
break-up within 4×10^4 yr	81 %	59 %	33 %	11 %	3.5 %	0 %

Against the break-ups within 4×10^4 yr (245 members):

- MA01 (ratio < 1) finds 98 % of them and flags 32 % of the members that do not break up. Its best threshold is 0.89, with an area under the ROC curve of 0.93.
- V22 (ratio < 1) finds 51 % and flags 6 %. Its best threshold is about 1.17.
- The neural network of Vynatheya et al. (2023) finds 25 %. Even for the label it was trained on (the ghost test, 198 unstable members) it finds only 32 %.
- These members lie just outside its training set: the mass ratio within binary A is 10.7, the training set has at most 10.

For a 2+2 with one widened, strongly unequal binary, MA01 is therefore the safe criterion: it hardly misses a break-up and errs on the side of instability. V22 at its nominal threshold misses half of them. The onset of CzeV343, MA01 = 1.00 and V22 = 1.19–1.24, lies where

5–11 % of such systems break up within 4×10^4 yr. This agrees with stage 2, where none of the 1800 members broke up in 10^4 yr.

Comparison with Pejcha et al. (2022). The two studies agree on three points:

- the instability, if it comes, is caused by binary A widening beyond about 50 $R_\odot$;
- most realisations of binary A end in a merger;
- binary B does not trigger the instability first.

They differ in how often A widens before it merges: $\gtrsim 85$ % in COMPAS, but only 10–29 % here. Here most histories reach contact while both stars are still on the main sequence. B does not widen at all here, while 60 % of the COMPAS realisations of B widen to 80–150 $R_\odot$. Both codes describe mass transfer with parametrised prescriptions, not with detailed stellar models. They differ in the stellar tracks (MIST here, SSE fitting formulae in COMPAS) and in how the start of contact and its consequences are treated.

The stellar tracks decide. To separate the two, the same 300 draws of the fiducial solution were run again with SSE tracks (the Hurley et al. 2000 fitting formulae on which COMPAS is built) instead of MIST, with everything else unchanged (**Table 2a**). With SSE, binary A leaves a stripped donor in 82 % of the draws, and its largest separation exceeds 50 $R_\odot$ in 83 % (median 65.6 $R_\odot$, against 7.3 $R_\odot$ with MIST). This is the widening that COMPAS finds in $\gtrsim 85$ % of its realisations. The outcome of the two track families agrees in only 77 of the 288 draws that finished with both (27 %). 137 draws that end in a common-envelope merger with MIST leave a stripped donor with SSE. With SSE the mass transfer starts later (median 70 Myr), and 75 % of the quadruples become dynamically unstable (1.25–2.00 Gyr, 16–84 %) against 14 % with MIST. The difference with COMPAS is therefore a difference of the stellar tracks, through the response of the donor and so the stability of its mass transfer, not of the dynamics. Which track family describes these stars better is a question for detailed binary-evolution models.

Table 2a. CzeV343, binary A: outcome of the same 300 draws (fiducial solution) with two families of stellar tracks.

outcome of binary A	MIST	SSE
common-envelope merger	59 %	11 %
contact	30 %	6 %
stripped donor, wide orbit	10 %	82 %
largest separation (median)	7.3 $R_\odot$	65.6 $R_\odot$
dynamically unstable (MA01)	14 %	75 %
onset of instability (16–50–84 %)	1.30–1.49–1.63 Gyr	1.25–1.65–2.00 Gyr

MIST: 288 draws finished, 12 stopped at the 300-s limit; the percentages refer to the finished draws.

An independent check is the grid of detailed case-A binary models by Nelson & Eggleton (2001). For initial mass ratios near 1.4 (here 1.76/1.27) and the shortest initial periods, their models typically “come into contact slowly, but reach contact before either star has left the main sequence” (their subtype AS). Their paper gives no separations, so it cannot say how far the orbit widens on the way. But contact on the main sequence is the outcome that dominates our histories, and it is the opposite of the widening that dominates in COMPAS. Our largest separations, 53–66 $R_\odot$, lie at the lower end of the COMPAS range. Whichever code is closer to the truth, the conclusion of this test is the same: when A widens, the quadruple crosses the MA01 limit at a separation of about 57–62 $R_\odot$.

2.4 The internal structure of the stripped donor

Before looking at stage 2 we estimate the apsidal constant of the donor, because the result depends on it.

At the onset the stellar track gives a helium core of 0.258 $M_\odot$ inside a star of 0.262 $M_\odot$. Only about 0.004 $M_\odot$, or 1.5 % of the mass, remains in the convective envelope, which reaches out to 14 $R_\odot$. We computed k_2 for such a star with a simple two-zone model (`tools/k2_condensed.py`):

- a compact core of radius 0.03 $R_\odot$;
- an envelope that is a polytrope of index n in the gravity of the core ($n = 1.5$ for a convective envelope, $n = 3$ for comparison).

The program integrates the Clairaut–Radau equation for the tidal distortion outwards through the star (Kopal 1959). The result, $k_2 = (3 - \eta_R)/(2(2 + \eta_R))$, follows from the logarithmic derivative η_R of the distortion at the surface. A uniform sphere gives the textbook 0.75, which checks the integration.

Table 2. Apsidal constant k_2 of a 0.26 $M_\odot$ star with a 0.258 $M_\odot$ core, a radius of 14 $R_\odot$ and envelopes of different mass. Columns:

- $M_{\rm env}$: mass of the envelope;
- $M_{\rm env}/M$: its fraction of the stellar mass;
- k_2 (n = 1.5): convective envelope;
- k_2 (n = 3): radiative-like envelope.

The first row is the state of the donor at the onset; the other rows show how k_2 grows with the envelope.

$M_{\rm env}$ [$M_\odot$]	$M_{\rm env}/M$	k_2 (n = 1.5)	k_2 (n = 3)
0.004	0.015	0.0006	0.00001
0.01	0.037	0.0015	0.00002
0.02	0.072	0.0028	0.00005
0.05	0.162	0.0065	0.00011
0.10	0.279	0.0116	0.00019

With the envelope the track predicts, $k_2 \approx 6 \times 10^{-4}$, even below the lowest value we integrated. To reach $k_2 = 0.01$ the star would need an envelope of about a quarter of its mass, which a stripped donor does not have. Observed stripped donors on their way to becoming helium white dwarfs, such as the pre-helium white dwarfs in EL CVn binaries, keep hydrogen envelopes of only about 0.001–0.005 $M_\odot$ (Maxted et al. 2014). We therefore regard $k_2 \approx 0.001$ **as the realistic case**, and 0.01 and 0.05 as deliberately generous upper bounds.

The model has two limits. It ignores the gravity of the envelope itself (≤ 1.5 % of the mass, a negligible error here) and the rotational flattening of the core. Both change k_2 by much less than the factor of ten between our three values.

2.5 Stage 2: the dynamics after the onset

Table 3. Outcome of stage 2: six onsets × 100 members, integrated for 10^4 yr, for each apsidal constant k_2 of the donor. Columns:

- *collided*: share of the 600 members in which the two stars of binary A collide;
- *collision time*: median of the six onsets (range of medians), and the 16–84 % range over all collisions;
- *largest* e_A: median and 84th percentile of the largest eccentricity that binary A reaches;
- *apsidal period*: period of the tidal and rotational apsidal motion of A at the onset, for draw fiducial 7.

k_2	collided	collision time after the onset [yr]	largest e_A (median, 84 %)	apsidal period [yr]
0.001	600 of 600	medians 59–201; all: 53–550	0.70, 0.72	589
0.01	0 of 600	–	0.16, 0.25	59
0.05	2 of 600	7200–8900	0.03, 0.05	12

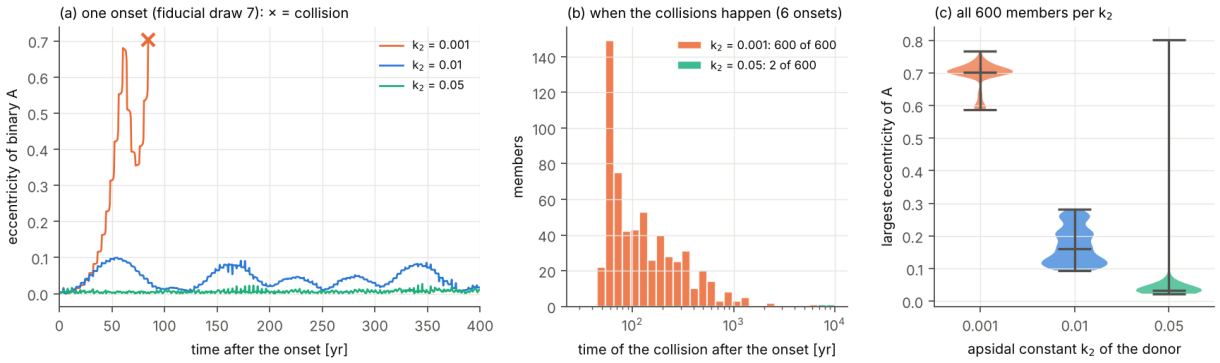


Figure 2. Stage 2 of CzeV343: the four-body motion after the onset of instability, for three values of the apsidal constant k_2 of the stripped donor (orange 0.001, blue 0.01, green 0.05). **(a)** Eccentricity of binary A over the first 400 years for one onset (fiducial draw 7, the same initial phases in all three runs). With k_2 = 0.001 the eccentricity climbs in steps to 0.7 and the two stars collide after 84 years (cross). With 0.01 it oscillates below 0.1, and with 0.05 it stays near zero. **(b)** When the collisions happen (years after the onset, logarithmic), all six onsets together; the legend gives how many of the 600 members collide. **(c)** Distribution of the largest eccentricity of A within 10^4 yr over all 600 members for each k_2 (violins: the width shows how many members have that value; bars: median, minimum and maximum).

Table 3 and Figure 2 show a sharp dichotomy.

With the realistic $k_2 = 0.001$ every member ends in a collision of the two stars of binary A. The collisions happen 53–550 years after the onset, with most of them within the first 150 years (panel b). The cause is visible in panel (a): the eccentricity of A grows from zero to about 0.7 within a few decades. This is the **von Zeipel–Lidov–Kozai (ZLK) mechanism**. A distant third body on an orbit strongly inclined to the inner binary exchanges angular momentum with it and periodically raises the inner eccentricity at the expense of the inclination. At the onset the mutual inclination of A and the outer orbit is 76–97° (medians of the six onsets), close to the 90° at which ZLK is strongest. Today the ZLK timescale of A is about 650 yr (Pejcha et al. 2022), much longer than its 33-yr apsidal period, so ZLK is quenched now; at the onset, with A eight times wider and its period 24 times longer (30 d instead of 1.2 d), the ZLK timescale, which scales as $P_{\rm out}^2/P_A$, has shrunk to about 650/24 ≈ 27 yr. The steps in the curve are the outer periastron passages: the outer orbit is so eccentric (0.70) that most of the perturbation acts during them.

At $e_A = 0.70$ the periastron distance of A is $a(1-e) \approx 0.3 \times 57.6 \approx 17\ \mathrm{R}_\odot$. That is just the sum of the two radii (14.0 + 3.7 $\mathrm{R}_\odot$), so the giant touches its companion and the integration records a collision. In reality this means a merger or a common envelope of the donor with its companion.

With $k_2 = 0.01$ or 0.05 the system survives 10^4 yr intact (598 of 600 members with 0.05). The eccentricity of A stays below 0.25 (0.01) or 0.05 (0.05). The reason is apsidal precession. ZLK works only if the orientation of the inner orbit stays coherent long enough for the third body to pump its eccentricity. Any additional apsidal precession faster than the ZLK cycle washes the effect out (Holman et al. 1997; Fabrycky & Tremaine 2007). The column *apsidal*

period in Table 3 shows the competing rates: the ZLK eccentricity grows within about 50–60 yr in panel (a).

- With $k_2 = 0.01$ the tidal and rotational bulges of the giant turn the orbit in 59 yr, about as fast as ZLK, so the eccentricity grows only to about 0.1–0.25 and oscillates (blue curve).
- With 0.05 the precession takes 12 yr and ZLK is fully quenched.
- With 0.001 it takes 589 yr, far too slow to matter.

General relativity adds a precession with a period of about 2.5×10^5 yr, irrelevant here.

The two late collisions with $k_2 = 0.05$ (after 7200 and 8900 yr, in one onset) come from a slow chaotic drift of the outer orbit and show that even the stiffest giant is not safe forever. They are 0.3 % of the members.

The quadruple is not torn apart. In none of the 1800 members is a star ejected, nor do the binaries exchange partners. The instability flagged by MA01 at the onset does not mean an immediate dynamical break-up. With V22 still above 1, the four-body configuration remains bound and hierarchical over 10^4 yr. What happens is slower and secular: the third body drives the eccentricity of the widened binary until its stars collide.

Numerical quality. The relative energy error stays below 7×10^{-8} for k_2 = 0.001 and 0.01, and below 1.4×10^{-6} for 0.05, where the stiffer bulges require more integration steps. Both are far below anything that could affect the outcome.

2.6 What the stage 2 model leaves out

- **Mass transfer during stage 2.** At the onset the donor still slightly overfills its Roche lobe (by 20–40 % in radius at zero eccentricity). The N-body integration keeps the masses fixed. Over the 50–550 yr until the collision the transfer moves only of order 10^{-6} $M_\odot$, so it cannot change the outcome. But it means the giant is in contact with its Roche lobe throughout, and the collision is in fact the onset of a common envelope.
- **Binary B as one body.** B is 16–20 times tighter than the outer periastron. Its internal structure affects the outer orbit only through its quadrupole moment, which is smaller than that of A by $(a_B/a_A)^2 \approx 10^{-2}$.
- **Tidal dissipation.** Tides are conservative here (no time lag). Dissipation would act on timescales of 10^4 yr or more for a 14 $R_\odot$ giant at 57 $R_\odot$ and could only circularise A between ZLK cycles. That matters for $k_2 \geq 0.01$, where the system survives, not for the realistic case.
- **Only six onsets.** They span the range of onset times and both solutions, and give the same result for each k_2 (panel b of Figure 2 combines them). We see no sign that more onsets would change the picture.

2.7 Is CzeV343 in the 3:2 resonance?

The periods of the two binaries, 1.2093 and 0.8069 d, are close to 3:2. A first-order resonance would show as a libration of one of the angles

$$\varphi_1 = 3\lambda_A - 2\lambda_B, \quad \varphi_2 = \varphi_1 - \varpi_A, \quad \varphi_3 = \varphi_1 - \varpi_B,$$

where λ is the mean longitude and ϖ the longitude of the periastron. In the present system all three angles circulate, with periods of 1.6–2.0 yr (400 yr of full physics). The present system is not in resonance.

How far is it from the resonance? 141 integrations of 1000 yr scanned the period of A from 1.20995 to 1.21030 d in steps of 2.5×10^{-6} d (energy error below 10^{-10}). The mean rate of each angle falls linearly with P_A, by 4702 rad/yr per day. For φ_1 and φ_3 it passes through zero without a plateau, at 1.2101625 and 1.2100224 d: these resonances, if present, are narrower than the step. For φ_2 the rate stays at zero over four steps, from 1.210125 to 1.2101325 d, instead of changing by ±0.017 rad/yr. There the angle librates with an amplitude of 66–330°

and does not complete a turn in 1000 yr. This is the 3:2 Lindblad resonance that Tremaine (2020) identifies in 2+2 quadruples, whose argument contains the periastron of the binary with the longer period, here A, at first order in its eccentricity. Its width is about 10^{-5} d in P_A, or 8×10^{-6} in the period ratio. CzeV343 lies 7.9×10^{-4} d, about 80 widths, below it.

The outer orbit splits the resonance into satellites at $3n_A - 2n_B - \dot{\varpi}_A + k\, n_{\rm out} = 0$, spaced by $0.47\,|k|\,\epsilon^{3/2}$ in $1 - 2P_A/3P_B$ (Tremaine 2020; $\epsilon \approx 0.007$ here), with a strength proportional to $e_{\rm out}^{|k|}$. With $e_{\rm out} = 0.70$ they are not weak. In the 400-yr run of the present system all the angles $\varphi_2 + k\lambda_{\rm out}$ with $k = -6 \ldots + 6$ circulate. The slowest, $k = -2$, has a period of 11.4 yr. Its period ratio near 3:2 is therefore not the result of a present resonance. Why so many 2+2 quadruples lie near such ratios (Kolář et al. 2025) remains open.

2.8 Answer

- In most futures of CzeV343 the quadruple does **not** fall apart dynamically at all. Binary A merges (fiducial solution, median 1.4 Gyr from now) or becomes a contact binary (alternative solution), and the system continues as a hierarchical system with one merged or contact pair.
- **Dynamical instability** occurs in 14 % of the fiducial and 1 % of the alternative draws, 1.3–1.6 Gyr (fiducial, 16–84 %) from now. In these draws the donor of A is stripped and the binary widens to 57–62 $R_\odot$.
- **What follows** depends on the internal structure of the stripped giant. With the envelope its track predicts ($k_2 \approx 10^{-3}$), ZLK cycles driven by binary B raise the eccentricity of A to 0.7. The giant collides with its companion within 50–550 years: a merger, or a common envelope inside a quadruple. Only if the giant kept a quarter of its mass in the envelope ($k_2 \gtrsim 0.01$) would its tidal bulge quench ZLK and keep the system together for at least 10^4 yr.
- No ejection or exchange of stars was found in any of the 1800 integrations.
- How often binary A widens depends on the stellar tracks: in 10 % of the draws with MIST, in 82 % with SSE on the same draws, which explains the difference from COMPAS.
- CzeV343 is not in the 3:2 resonance; it lies about 80 resonance widths away.

The prediction is testable in principle on other systems: a stripped red giant in a 2+2 quadruple with a strongly inclined outer orbit should be rare, because it would not survive long.

Gaia note. The Gaia DR3 astrometric solution (Gaia Collaboration 2023) of CzeV343 has a renormalised unit weight error (RUWE) of 1.62. Values above about 1.4 indicate that a single-star model does not fit the motion of the light centre (Lindegren et al. 2018). For CzeV343 this is consistent with the 4-yr mutual orbit of the two binaries at the Gaia parallax of 0.224 ± 0.025 mas. We did not use it quantitatively.

3 CzeV4315: a 2+2 quadruple or a 2+(2+1) quintuple?

3.1 The system and the question

CzeV4315 (HD 228777) contains two eclipsing binaries (Zasche et al. 2025):

- **Binary A**: period 6.73907 d, eccentricity 0.64, inclination 87.44 ± 0.21°. It has eclipsed steadily for as long as it has been observed.
- **Binary B**: period 0.9193238 d, circular. Its inclination grew from below 57.4° to 63.42 ± 0.25° within a few years, so its eclipses deepen at a rate of about 1.4° per year. The times of its eclipses also show a **119-day periodicity**.

The inclination of B was below 57.4° in TESS sector 15 (2019) and grew through 58.6° (2021), 60.3° (2022) and 62.2° (2024) to 62.9° (TESS sector 81, 2024) (Zasche et al. 2025, their Table 3). The paper gives no amplitude of the 119-day signal and no spectroscopy.

Zasche et al. left open how the two binaries are arranged; they called the two possibilities "2+2+1" and "2+1+2". They noted one weak hint in favour of the second: A shows no fast

apsidal motion, which a strong dynamical interaction with B would cause. Two arrangements explain the 119-day signal:

- **2+2**: A and B orbit each other with a period of 119 d. The 119-day signal is the light-travel time and the dynamical perturbation of this mutual orbit.
- **2+(2+1)** (the "2+1+2" of Zasche et al.): B has its own third star (m5) on a 119-day orbit (called Bt below). This triple B+m5 then orbits binary A on a much wider, unknown orbit.

A fast change of inclination needs a strong perturber close to B. The question is which arrangement can turn B by 1.4°/yr **without** turning A as well, and without destroying the system.

Masses and radii. These are estimates made here, not values from the paper. Each binary was assumed to have a primary of about 10 000 K (spectral type B9 V); the masses follow from the measured mass ratios and the radii from the relative radii R/a. The values are 2.40 + 2.09 $M_\odot$ for A and 2.30 + 2.35 $M_\odot$ for B, with radii of 1.6–1.8 $R_\odot$.

3.2 Method: the same data, two hierarchies

We built one ensemble for each hierarchy and weighted every member by the same observations:

1. **Inclination change of B.** The rate of change of $|i_B - 90°|$ today must be −1.4 ± 0.2°/yr, which enters as a Gaussian likelihood. The negative sign means that B turns towards edge-on.
2. **A keeps eclipsing.** Integrated backwards, A must have eclipsed for at least the last 5 years. Integrated forwards, it must keep eclipsing for at least 3 months. Members violating either condition get zero weight.
3. **No past mass transfer.** No star may have filled its Roche lobe during the backward integration, because both binaries are detached.
4. **Survival.** The system must be intact in both directions of time. A member that collides or loses a star within the integrated span cannot be the system we see.

Table 4. The two ensembles. Columns:

- *drawn parameters*: the unknown quantities drawn from priors;
- *physics*: the forces included;
- *span*: the integration time backwards and forwards;
- *members*: the number of integrated members.

hierarchy	drawn parameters (priors)	physics	span	members
2+2	$P_{\rm out}$ = 119 ± 3 d, $e_{\rm out}$ 0–0.7, orientation of the mutual orbit (isotropic); node and sense of rotation of B; i_A, ω_A	N-body, GR, tides, rotation	±100 yr	1000

hierarchy	drawn parameters (priors)	physics	span	members
2+(2+1)	m_5 0.3–6 $M_\odot$ (log-uniform); P_{Bt} = 119 ± 3 d, e_{Bt} 0–0.8, orientation of Bt (isotropic); P_{out} 1500–100 000 d (log-uniform), e_{out} 0–0.7, orientation (isotropic); node of B; i_A, ω_A	N-body, GR	±60 yr	1500

Both ensembles keep only members that are stable by the criterion of Vynatheya et al. (2022) at the start. The 2+2 ensemble includes tides and rotation because its 119-day mutual orbit is tight. In the quintuple the tidal effects on the orbital planes are negligible over 60 years compared with the perturbation by m5. The shorter span of the quintuple is still longer than any of the constraints, which reach at most 5 years back.

3.3 Results

Table 5. Results of the two ensembles. Columns:

- *matching tilt*: members whose rate of B lies within ±2σ of the observed value;
- *of these: A keeps eclipsing*: how many of those also keep A eclipsing and stay intact;
- *tilt rate of A*: the median rate of change of the inclination of A in the members that match B (the observed rate is close to zero);
- *evidence Z*: the mean observation weight over the ensemble;
- n_{eff}: effective number of members carrying the result.

hierarchy	matching tilt	of these: A keeps eclipsing, intact	tilt rate of A [°/yr] (median)	evidence Z	n_{eff}
2+2	141 of 1000	0	2–50 (typical 10–20)	3×10^{-16}	1
2+(2+1)	80 of 1500	62 intact, A eclipsing	0.001 (all < 0.22)	0.022	46

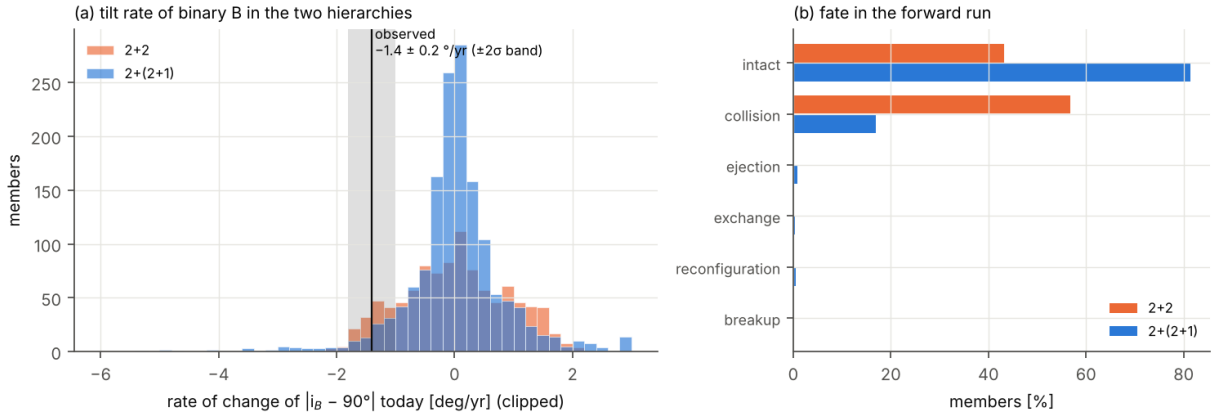


Figure 3. CzeV4315 in the two hierarchies (orange 2+2, blue 2+(2+1)). **(a)** Histogram of the present rate of change of $|i_B - 90°|$ of binary B in all members (degrees per year, values outside −6…+3 clipped to the edge). The black line and grey band mark the observed −1.4 ± 0.2°/yr and its ±2σ range. **(b)** What happens to the members in the forward integration: intact, collision of two stars, ejection of a star, exchange of partners, reconfiguration of the hierarchy, or break-up.

Both hierarchies can turn B fast enough. In Figure 3a both histograms reach the observed band, so the rate of B alone does not decide.

In a 2+2 the same perturbation also turns A. The 141 members of the 2+2 ensemble that match the rate of B fall into two groups:

- 66 of them end in a collision, of the two stars of B or of a star of B with A. At a period of 119 d the mutual orbit lies close to the stability limit, and an eccentric mutual orbit drives the stars together. This is why 57 % of all 2+2 members collide (panel b).
- The other 75 survive, but in all of them A tilts by 2–50° per year, typically 10–20°. A would stop eclipsing within months (median 0.4 yr), whereas it has eclipsed for years.

The reason is simple. In a 2+2 with a 119-day mutual orbit, A is the third body of B and B is the third body of A, at the same distance. A perturbation strong enough to turn B by 1.4°/yr turns A too: A's orbit is wider (6.7 d against 0.9 d) and therefore even more sensitive. No 2+2 member satisfies all the observations. The four non-zero weights are below 3×10^{-13}.

In a 2+(2+1) the perturber acts on B alone. The third star m5 sits on its 119-day orbit around B only. This is also the arrangement for which Zasche et al. saw a "weak indirect hint": A shows no fast apsidal motion. They left the question open. A is far away, at 4000–50 000 d (16–84 %) in the weighted members, and its inclination changes by less than 0.22°/yr in every matching member. 81 % of all members and 99 % of the weight remain intact (panel b).

Evidence. The mean weights are $Z = 0.022$ for 2+(2+1), carried by 46 effective members, and 3×10^{-16} for 2+2. The 2+2 value rests on effectively one member and is therefore only an upper estimate. A safer statement is this: with 1000 members and no member giving a weight above 10^{-12}, the evidence of the 2+2 is below about 10^{-3}, otherwise at least one member would have matched. The Bayes factor in favour of 2+(2+1) is therefore at least about 20 by this conservative bound, and formally 10^{13}–10^{14}. Either way the data clearly prefer a third star around B.

The comparison has two caveats. The quintuple ensemble has no tides and a shorter span than the 2+2 ensemble (Table 4). Requiring survival changes its evidence by less than 1 % (0.0216 instead of 0.0218), and tides cannot rescue the 2+2, whose problem is the tilt of A, not survival.

3.4 What the data say about the third star

Table 6. Weighted posterior of the 2+(2+1) hierarchy. Each row gives the 16th, 50th and 84th percentiles of a parameter over the members, weighted by their observation weights.

The prior range is in the last column for comparison; a posterior that fills its prior is not constrained by the data.

parameter	16 %	50 %	84 %	prior
mass of the third star m_5 [$M_\odot$]	1.4	3.2	4.9	0.3–6, log-uniform
period of its orbit P_{Bt} [d]	116.0	118.4	120.9	119 ± 3
eccentricity of its orbit e_{Bt}	0.18	0.46	0.73	0–0.8
mutual inclination of B and Bt [°]	31	54	141	isotropic
period of A around the triple P_{out} [d]	4100	16 600	52 300	1500–100 000

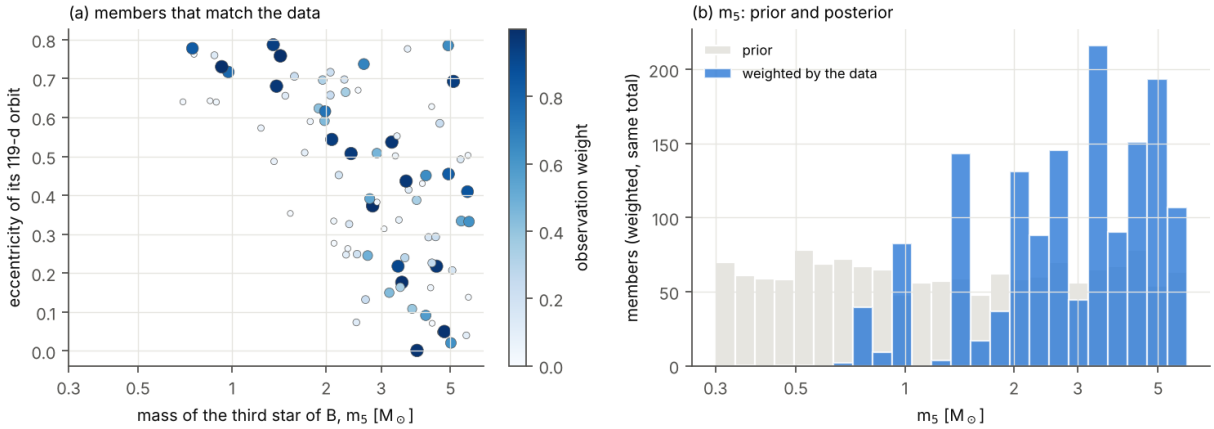


Figure 4. The third star of binary B in the 2+(2+1) ensemble. **(a)** Mass m_5 (logarithmic axis) against the eccentricity of its 119-day orbit, for every member with an observation weight above 1 % of the largest one. Colour and size of each point give its weight (dark and large = good match). **(b)** Distribution of m_5: the prior (grey, all 1500 members) and the same members weighted by the data (blue, scaled to the same total). Where the blue bars exceed the grey ones the data favour that mass.

The mass of the third star alone is not determined. In Figure 4b the weighted distribution rises above the prior only for $m_5 \gtrsim 1$ $M_\odot$ and then fills the prior up to its upper edge at 6 $M_\odot$. 16–84 % lies at 1.4–4.9 $M_\odot$, and 95 % at 0.75–5.7 $M_\odot$. Values below about 0.7 $M_\odot$ are excluded: a light star cannot turn B fast enough.

Panel (a) shows why the upper side stays open. The rate at which B's plane precedes grows with the mass of the perturber and with $(1-e_{Bt}^2)^{-3/2}$, and it depends on the mutual inclination through $\cos i_{mut}$. A light star on an eccentric orbit (upper left) and a heavy one on a nearly circular orbit (lower right) produce the same rate. The data constrain this combination, not the mass. A first ensemble with the mass limited to 3 $M_\odot$ seemed to place m_5 near 2–2.8 $M_\odot$. That was the edge of the prior, not a measurement, which is why the prior was widened here.

This makes a testable prediction. 74 % of the weight lies at $m_5 > 2$ $M_\odot$. Such a star is about as massive as the stars of B (2.3 $M_\odot$) and, if on the main sequence, about as bright. It would show up as:

- third light in the light curve of B;

- a third set of spectral lines;
- a light-travel-time effect in the eclipse times of B. For the median posterior (m5 = 3.2 $M_\odot$) the semi-major axis of the 119-day orbit is about 0.94 au and B moves on 0.38 au around the centre of mass, which gives up to about 3 minutes, times sin of the inclination of Bt. The 119-day timing signal of Zasche et al. can therefore be checked against this once its amplitude is measured. Radial velocities of B would vary over 119 d with a semi-amplitude of up to about 35 km/s.

A spectrum that shows only the four stars of A and B would push m_5 to the lower end, about 1 $M_\odot$, and therefore to a high eccentricity of Bt (Figure 4a, upper left).

Prediction for the eclipses.

- In the weighted members B stops or changes its eclipsing regime within about 30–60 years (16–50 %); in the remaining third nothing changes within the 60 years integrated.
- A keeps eclipsing in all weighted members over the whole span.

3.5 The fifth star seen by Gaia

Zasche et al. (2025) noted that CzeV4315 is a visual double with a separation of about 1.9″ and almost no relative motion in the Washington Double Star Catalog, which suggests an orbital period of thousands of years. Gaia DR3 (Gaia Collaboration 2023) resolves this companion 1.84″ from CzeV4315. It has G = 11.92 (CzeV4315 itself has G = 9.65) and a parallax of 0.826 ± 0.036 mas, equal within the errors to that of CzeV4315 (0.893 ± 0.014 mas, about 1.1 kpc). It is therefore a physical companion at a projected distance of about 2000 au. Being 2.3 mag fainter, it cannot be one of the binaries, whose stars are similar. It is also too far away to be the 119-day star m5. It is another member of the system, which makes CzeV4315 at least a sextuple in the 2+(2+1) picture. It does not affect the dynamics on the timescales studied here. Being 2.3 mag fainter, it adds about 11 % to the light of every photometric measurement that does not resolve it (TESS pixels are 21″), which is part of the large third light in the TESS fits.

4 Are the posteriors converged? A 1000-step MCMC

4.1 The question

The uncertainties quoted in Parts II and III come from Markov-chain Monte Carlo (MCMC) sampling of the posterior probability of the fitted parameters. We use the affine-invariant ensemble sampler emcee (Foreman-Mackey et al. 2013); the fit chain is described in *Physics and Methods*, chapter Fitting observed data. An MCMC run is a set of **walkers** that move through parameter space. Each **step** costs one full N-body integration per walker. After a **burn-in** the positions of the walkers sample the posterior.

The positions of one walker are correlated from step to step. The **integrated autocorrelation time** τ is the number of steps after which a walker has "forgotten" where it was. A chain of N steps contains roughly N/τ independent samples per walker. Foreman-Mackey et al. (2013) suggest running "a few (say 10) autocorrelation times"; the emcee documentation recommends at least 50 τ before the estimate of τ itself is reliable. SWINGERsim discards the first half of the chain as burn-in, measures τ on the second half, and warns when the kept half is shorter than 10 τ.

The first fits used 400 steps, for which emcee reported $\tau \approx 21$ (19 τ). To test whether the quoted uncertainties are reliable, we re-ran three fits with 1000 steps: one 2+2 system (BU CMi), one 3+1 system (TIC 120362137) and one 2+1 system with a changing inclination (V907 Sco).

4.2 Results

Table 7. The 1000-step runs. Columns:

- *run time*: wall-clock time on 12 cores;

- τ: the mean integrated autocorrelation time over the parameters, in steps;
- *steps/τ*: the chain length in units of τ;
- *error ratio*: 1σ uncertainties from 1000 steps divided by those from 400 steps, as the median over all parameters and the range;
- *largest shift*: the largest change of a median value between the two runs, in units of its uncertainty.

system	run time	τ [steps]	steps/τ	error ratio (median; range)	largest shift
TIC 120362137	10.5 h	53.3	18.7	1.05; 0.86–2.02	< 0.4σ
V907 Sco	19.3 h	53.9	18.6	1.10; 0.86–1.53	< 0.8σ
BU CMi	19.4 h	52.3	19.1	1.03; 0.87–1.36	< 0.25σ

Three things stand out.

1. **The autocorrelation time was underestimated.** It grew from about 21 to about 53 steps when the chain was made 2.5 times longer. This is the known behaviour of short chains: τ cannot be measured from a chain shorter than a few tens of τ, and the estimate grows with the chain (Foreman-Mackey et al. 2013; Sokal 1997). The chains are therefore about 19 τ long in total, and the half kept after the burn-in is about 9.4 τ. That is just below the 10 τ of Foreman-Mackey et al. and well short of the 50 τ of the emcee documentation.
2. **The results did not move.** No median moved by more than 0.8σ, and most by much less. The typical uncertainty grew by 3–10 %. In a few weakly constrained parameters with broad, non-Gaussian posteriors it grew by up to a factor of two (TIC 120362137) or 1.5 (V907 Sco).
3. **MCMC and linearised uncertainties agree.** The covariance matrix of the least-squares fit (the “linear” errors) agrees with the MCMC widths to about 10 % in the median. Both describe the same well-defined minimum.

4.3 What this means for the quoted numbers

- The **medians** quoted in Parts II and III are robust. Doubling the chain length would not move them by more than a fraction of their uncertainty.
- The **1σ uncertainties** are reliable to a few tens of per cent. They are slightly too small rather than too large, by about 5–10 % typically and more for the weakly constrained elements.
- The **tails** (95 % ranges) are less certain than the 1σ widths. With about 9–10 independent samples per walker they rest on fewer samples. This matters most for the dates of future events quoted as 95 % ranges, such as the end of the eclipses of V907 Sco in 2052.5–2056.8. The width of such ranges is itself uncertain, more than that of the 1σ intervals.
- Ten τ in the kept half would need about 1100 steps, just above the 1000 used. The 50 τ of the emcee documentation would need about 5300 steps, i.e. 55–100 h per system on 12 cores. The comparison of 400 and 1000 steps shows that longer chains refine the widths but do not change the conclusions. We report the 1000-step results.

The values in Parts II and III are those of the 1000-step runs.

5 Conclusions

Part IV used SWINGERsim on three questions that the literature leaves open.

1. **CzeV343.**
   - The quadruple becomes dynamically unstable only if the donor of binary A is stripped and the binary widens to about 57–62 $R_\odot$. With the mass-transfer prescriptions of SWINGERsim this happens in 14 % of the futures of the fiducial solution (1.3–1.6 Gyr from now) and in 1 % of the alternative solution; in most futures binary A merges or becomes a contact binary first. How often A widens depends on the stellar tracks: the same draws with SSE instead of MIST widen it in 82 % (COMPAS, built on the same formulae: $\gtrsim$ 85 %; Pejcha et al. 2022), and 75 % of them become unstable. The onset separation and what follows do not depend on this.
   - Near the onset the Mardling–Aarseth criterion is the safe one for such systems: in 1200 N-body realisations it finds 98 % of the break-ups within 4×10^4 yr, the criterion of Vynatheya et al. (2022) 51 % and their neural network 25 %.
   - CzeV343 is not in the 3:2 resonance between its binaries; it lies about 80 resonance widths away.
   - After the onset the system is not torn apart. With the apsidal constant computed for the stripped giant ($k_2 \approx 10^{-3}$), ZLK cycles driven by binary B raise the eccentricity of A to 0.7, and the giant collides with its companion within 50–550 years in all 600 integrations.
   - Only an implausibly large envelope ($k_2 \gtrsim 0.01$) would quench ZLK and keep the system intact for 10^4 yr.
   - The answer to "when and how" is therefore: if binary A widens (rarely in our models, often in COMPAS), the instability starts at 57–62 $R_\odot$, in our models about 1.3–1.6 Gyr from now, and ends within centuries in a merger inside the quadruple driven by ZLK, not in an ejection.
2. **CzeV4315.**
   - The fast inclination change of binary B cannot come from binary A. In a 2+2 the perturbation that turns B also turns A by 2–50°/yr, or the system collides.
   - A third star on the 119-day orbit around B explains all the observations with an evidence larger by a factor of at least 20 (formally 10^{13}). Zasche et al. (2025) left the arrangement open, with a weak hint in the same direction.
   - Its mass is degenerate with the eccentricity and inclination of its orbit. 74 % of the weight lies above 2 $M_\odot$, which spectroscopy or the light-travel-time amplitude of the 119-day signal can test.
   - A further physical companion at 1.84″ makes the system at least sextuple.
3. **Convergence of the fits.**
   - The 1000-step MCMC runs are about 19 autocorrelation times long; the half kept after the burn-in is about 9.4 τ.
   - Their medians agree with the 400-step runs within 0.8σ, and their uncertainties are larger by 3–10 % typically.
   - The values quoted in Parts II and III are robust. Their 1σ uncertainties are reliable to a few tens of per cent, and their 95 % ranges somewhat less.

What these tests show about the software. Each question needed several parts of SWINGERsim working together:

- stellar evolution and mass transfer with stability tracking (CzeV343, stage 1);
- N-body integration with relativity, tides and rotation, set up automatically from an evolved state (stage 2);
- ensembles with observation weights and evidences, across two different hierarchies (CzeV4315);
- the fit chain with MCMC (Section 4).

The answers depend on physical inputs that the observations do not fix: the apsidal constant of a stripped giant, the mass of an unseen star. SWINGERsim makes these dependences explicit and quantifies them instead of hiding them in a single run.

Open points, collected in *Open questions and future work*:

- a detailed binary-evolution code on the CzeV343 inputs, to decide between the MIST and SSE behaviour of the donor of binary A;

- spectroscopy of CzeV4315, to detect or exclude a massive third star around B, and a measurement of the amplitude of its 119-day timing signal;
- longer MCMC chains (≈ 1100 steps for 10 τ after the burn-in, ≈ 5300 for 50 τ) where the tails of a posterior matter.

6 Data and reproducibility

Table 8. Inputs and outputs of Part IV. Each row names a configuration or tool (input) and the output folder it produces; paths are relative to the SWINGERsim directory.

item	input	output
CzeV343 stage 1 and stage-2 set-ups	`tools/czev343_future.py`, `configs/special/czev343.yaml`	`output/special/czev343/stage1.csv`, `stage2/*.yaml`
CzeV343 stage 2	`swingersim ensemble output/special/czev343/stage2/<onset>_k2_<k2>.yaml`	`output/special/czev343/s2tides/`
apsidal constant of the stripped giant	`tools/k2_condensed.py`	Table 2
CzeV4315, 2+2	`configs/special/czev4315_2p2_nbody_ensemble.yaml`	`output/special/czev4315_2p2/`
CzeV4315, 2+(2+1)	`configs/squadra/czev4315_2p2p1_wide_ensemble.yaml`	`output/special/czev4315_2p2p1_wide/`
1000-step MCMC	fit configurations of Parts II and III with 1000 steps	the fit folders of Parts II and III
figures and numbers	`tools/part4_figures.py`	`docs/figures/part4/`

The commands are:

```
python tools/czev343_future.py -o output/special/czev343          # stage 1 + stage-2 set-ups
for f in output/special/czev343/stage2/*_k2_*.yaml; do
  swingersim ensemble "$f" -o "output/special/czev343/s2tides/$(basename "$f" .yaml)" -j 12
done
swingersim ensemble configs/special/czev4315_2p2_nbody_ensemble.yaml -o
 ↪  output/special/czev4315_2p2 -j 12
swingersim ensemble configs/squadra/czev4315_2p2p1_wide_ensemble.yaml -o
 ↪  output/special/czev4315_2p2p1_wide -j 12
python tools/k2_condensed.py
python tools/part4_figures.py
```

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